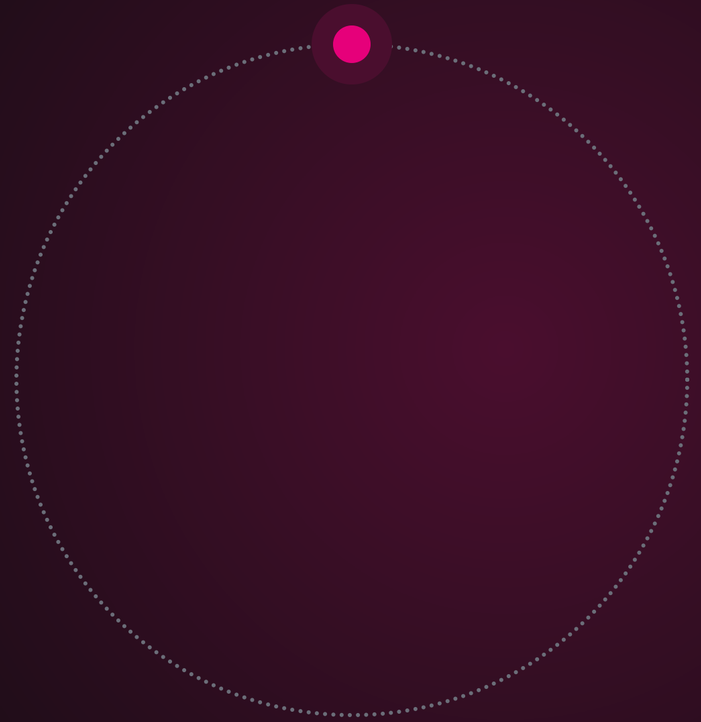
POLKADOT ASSEMBLE · MUMBAI Sat 7 Nov 2026 · Main stage

The Backend You Don’t Run

Private messaging, storage and personhood with **Polkadot Trinity**



Dhiraj Parity Technologies

THE PROBLEM

Count the servers

6 servers in a typical 2026 dApp

Frontend host

DNS hijack · Apr 2026

RPC provider

Cut off Venezuela · 2022

IPFS gateway

ipfs.io sunset · Sep 2026

Chat backend

Hosted network ends · 2027

Signaling server

One more box to run

Bot filter / KYC

Orb paused in India · 2023

Ethereum has the building blocks. **The gap is the defaults.**

THE IDEA · POLKADOT TRINITY

What if the backend were the network?

...and you run no servers to do it.



servers you run

“Less trust, more truth.”

Shawn Tabrizi, The Future of Web3

THE STACK

The shape of the stack

YOUR APP	A sandboxed app at name.dot

↓ talks to the host through one interface: TrUAPI

THE HOST	Mobile, desktop or web (dot.li)

↓ web: smoldot light client by default

POLKADOT	Asset Hub · People · Bulletin

Polkadot Trinity: Celerity (messages) · Levity (storage) · Humanity (personhood)

THE STACK

The light client is the connection

MOST APPS TODAY

Your app → one RPC server

You trust it.

TRINITY, ON THE WEB

Your app → light client → peers

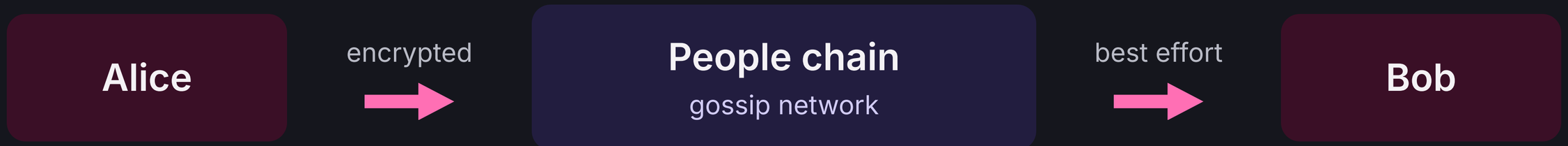
You check it.

No major Ethereum wallet ships a light client by default. Defaults decide.

BUILDING BLOCK 1 · CELERITY

Private messages, no server

Live on People chain | Apps on testnet



The app encrypts, not the network

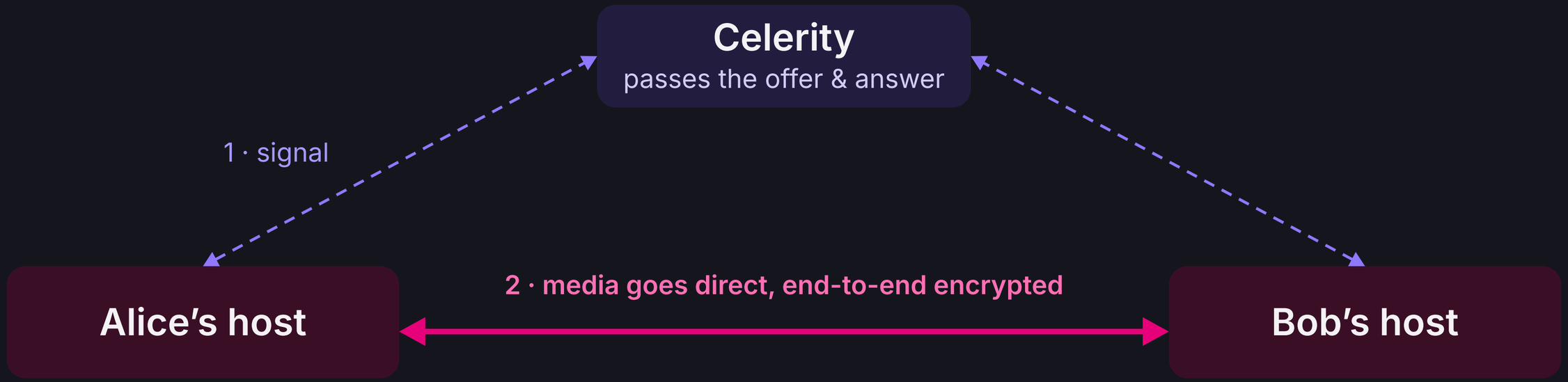
A topic only the two of you know

Every message expires

...and you run no servers to do it.

BUILDING BLOCK 1 · CELERITY

Calls with no signaling server



If a direct path fails, a TURN relay helps, as in any WebRTC app.

BUILDING BLOCK 2 · LEVITY

Storage that expires, on purpose

Live on Polkadot · 1 Sep 2026

App publishing on testnet

14

days, renewable

- A hash (CID) names the data
- Fetched by CID, checked on the web
- Whole apps on .dot names

Like Ethereum blobs (about 18 days), but you fetch it by CID. Need permanence? Mirror to Arweave.

...and you run no servers to do it.

BUILDING BLOCK 3 · HUMANITY

Prove you’re a person, not which one

Pallets live on People chain | The game is on testnet



A weekly game in groups of six.

No iris scan, no state ID.

One alias per app.

Apps can’t link you.

THE TWIST

Personhood is the rate limiter

Quotas per person, not per wallet.

WAKU	XMTP	TRINITY
by stake	by fees	by person

PUT TOGETHER · IMAGINE AN APP FOR THIS EVENT

One alias per person

HUMANITY
One alias each

CELERITY
Private chat

LEVITY
Schedule & site

WEB LIGHT CLIENT
Checked reads

ONE APP

assemble.dot

No backend to run.
No infra team.
No server bill.

...and you run no servers to do it.

HONESTLY

Where it stands today

Live on Polkadot

- Bulletin chain
- Statement store
- Personhood pallets

On testnet

- Developer kit
- Personhood game
- App publishing

Build now. Name the layer that’s live.

No public user numbers yet.

START · ON TESTNET

Start today, in three steps

1

RUN IT LOCALLY

```
$ truapi-host dev -- yarn dev
```

2

ADD THE SDK

```
$ yarn add @parity/product-sdk
```

3

SHIP TO A .DOT NAME

```
$ bulletin-deploy --env paseo-next-v2 ./out you.paseo
```

Scan for the quick start

Count your servers.
Then delete them.

github.com/paritytech

docs.polkadot.com/apps

Thank you. Questions?